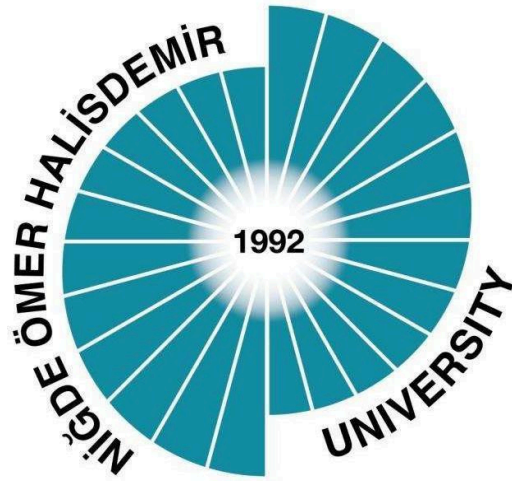


T.R.

NİĞDE ÖMER HALİSDEMİR UNIVERSITY

FACULTY OF AGRICULTURAL SCIENCES AND TECHNOLOGIES

DEPARTMENT OF BIOSYSTEMS ENGINEERING



DEVELOPMENT OF AN IoT-BASED SMART WEARABLE DEVICE FOR
MONITORING THE HEALTH AND ACTIVITY OF FARM ANIMALS

ABASS ISSAKA MOHAMMED

LÜTFULLAH METİN ÜNLÜ

Graduation Thesis

Supervisor: Dr. Mehmet Kürşat YALÇIN

JUNE 2026

THESIS CERTIFICATION

It is certified that this thesis has been fully written by ABASS ISSAKA MOHAMMED AND LÜTFULLAH METİN ÜNLÜ, providing all the necessary information. It is also certified that all information and sources used in this thesis were cited and referenced definitively, by the author.

ABASS ISSAKA MOHAMMED
LÜTFULLAH METİN ÜNLÜ

SUMMARY

**DEVELOPMENT OF AN IoT-BASED SMART WEARABLE DEVICE FOR
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SUPERVISOR: Dr. Mehmet Kürşat YALÇIN

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The rapid advancement of the Internet of Things (IoT) and smart sensing technologies has created new opportunities for improving livestock management and animal welfare in modern agriculture. This study presents the development of an IoT-based smart wearable device for real-time monitoring of the health and activity of farm animals. The proposed system was designed to assist farmers and herdsman in tracking animal movement, behavior, and environmental conditions remotely, thereby improving farm management efficiency, animal welfare, and early disease detection.

The developed system consists of two main units: an animal collar node and a herdsman gateway node. The animal collar node integrates an ESP32 microcontroller, MPU9250 motion sensor, thermistor temperature sensor, rechargeable battery, and wireless communication capability based on ESP-MESH technology. The herdsman gateway node incorporates an ESP32 microcontroller, GPS module, LoRa communication module, and rechargeable power supply. Communication between collar nodes and the gateway was achieved using ESP-MESH networking, while long-range transmission from the gateway to the monitoring station was implemented using LoRa technology.

The MPU9250 sensor was utilized to monitor livestock activity and movement through accelerometer and gyroscope measurements, while the thermistor sensor was used to estimate body temperature conditions. GPS technology enabled location tracking within the livestock monitoring environment. Sensor data collected from the collar nodes were transmitted through the mesh network to the gateway node and subsequently forwarded over LoRa for remote monitoring and visualization.

The developed prototype successfully demonstrated the ability to acquire, process, and wirelessly transmit real-time livestock health and activity data. The proposed system provides a low-cost, scalable, and energy-efficient solution for smart livestock farming, with potential applications in precision agriculture, animal welfare monitoring, disease prevention, and farm security. This research contributes to the growing field of IoT-based agricultural systems by presenting an integrated wearable monitoring platform suitable for modern livestock management practices.

Keywords: IoT, ESP32, Livestock Monitoring, ESP-MESH, LoRa, Precision Agriculture

ÖZET

ÇİFTLİK HAYVANLARININ SAĞLIK VE AKTİVİTELERİNİ İZLEMEK İÇİN İOT
TABANLI AKILLI GİYİLEBİLİR CİHAZ GELİŞTİRİLMESİ

ABASS ISSAKA MOHAMMED AND LÜTFULLAH

METİN ÜNLÜ

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TARIMSAL BİLİMLER VE TEKNOLOJİLER FAKÜLTESİ

BİYOSİSTEM MÜHENDİSLİĞİ BÖLÜMÜ

DANIŞMAN: Dr. Mehmet Kürşat YALÇIN

HAZİRAN 2026, 37 sayfa

Nesnelerin İnterneti (IoT) ve akıllı algılama teknolojilerindeki hızlı gelişmeler, modern tarımda hayvancılık yönetimi ve hayvan refahının iyileştirilmesi için yeni fırsatlar yaratmıştır. Bu çalışma, çiftlik hayvanlarının sağlık ve aktivite durumlarının gerçek zamanlı olarak izlenmesi amacıyla IoT tabanlı akıllı giyilebilir bir cihazın geliştirilmesini sunmaktadır. Önerilen sistem, hayvan hareketlerini, davranışlarını ve çevresel koşulları uzaktan takip ederek çiftçilere ve çobanlara destek olmak, böylece çiftlik yönetim verimliliğini artırmak, hayvan refahını geliştirmek ve hastalıkların erken teşhisini sağlamak amacıyla tasarlanmıştır.

Geliştirilen sistem iki ana birimden oluşmaktadır: hayvan tasma düğümü ve çoban ağ geçidi düğümü. Hayvan tasma düğümü; ESP32 mikrodenetleyici, MPU9250 hareket sensörü, termistör sıcaklık sensörü, şarj edilebilir pil ve ESP-MESH teknolojisine dayalı kablosuz iletişim özelliğini içermektedir. Çoban ağ geçidi düğümü ise ESP32 mikrodenetleyici, GPS modülü, LoRa haberleşme modülü ve şarj edilebilir güç kaynağından oluşmaktadır. Tasma düğümleri ile ağ geçidi arasındaki iletişim ESP-MESH ağı kullanılarak sağlanırken, ağ geçidinden izleme istasyonuna uzun mesafeli veri iletimi LoRa teknolojisi ile gerçekleştirilmiştir.

MPU9250 sensörü, ivmeölçer ve jiroskop ölçümleri aracılığıyla hayvanların aktivite ve hareketlerini izlemek için kullanılmıştır. Termistör sensörü ise vücut sıcaklığı durumlarının tahmin edilmesinde kullanılmıştır. GPS teknolojisi, hayvancılık izleme ortamında konum takibini mümkün kılmıştır. Tasma düğümlerinden toplanan sensör verileri, mesh ağı üzerinden ağ geçidi düğümüne iletilmiş ve ardından uzaktan izleme ve görselleştirme amacıyla LoRa üzerinden aktarılmıştır.

Geliştirilen prototip, gerçek zamanlı hayvan sağlığı ve aktivite verilerini toplama, işleme ve kablosuz olarak iletme yeteneğini başarıyla göstermiştir. Önerilen sistem; hassas tarım, hayvan refahının izlenmesi, hastalıkların önlenmesi ve çiftlik güvenliği gibi alanlarda potansiyel uygulamalara sahip, düşük maliyetli, ölçeklenebilir ve enerji verimli bir çözüm sunmaktadır. Bu çalışma, modern hayvancılık yönetimi uygulamalarına uygun entegre bir giyilebilir izleme platformu sunarak IoT tabanlı tarımsal sistemler alanına katkı sağlamaktadır.

Anahtar Kelimeler: IoT, ESP32, Hayvancılık İzleme, ESP-MESH, LoRa, Hassas Tarım.

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SYMBOLS AND ABBREVIATION

Symbols	Description
°C	Degrees Celsius
mm	Millimeter
km	Kilometer
V	Voltage
A	Current
MHz	Megahertz

Abbreviations	Description
IoT	Internet of Things
GPS	Global Positioning System
MPU	Motion Processing Sensor
ESP-MESH	ESP32 Mesh Networking
LoRa	Long Range Communication
PCB	Printed Circuit Board
UART	Universal Asynchronous Receiver Transmitter
AI	Artificial Intelligence

CHAPTER I

INTRODUCTION

Livestock production plays a significant role in ensuring food security, economic development, and sustainable agricultural practices worldwide. Farm animals such as cattle, sheep, and goats provide essential products including meat, milk, leather, and other by-products that contribute substantially to the agricultural economy. However, effective livestock management remains a major challenge, particularly in extensive grazing systems where animals are dispersed over large geographical areas and continuous monitoring is difficult. Traditional livestock management practices rely heavily on visual observation by farmers and herdsman, making the detection of health problems, abnormal behavior, and movement patterns time-consuming and inefficient. As livestock populations increase and farming systems become more complex, there is a growing need for intelligent technologies capable of supporting real-time monitoring and decision-making in animal production systems.

Recent advancements in the Internet of Things (IoT), wireless communication, embedded systems, and wearable sensing technologies have created new opportunities for precision livestock farming. Precision livestock farming focuses on the continuous monitoring of animals through automated technologies to improve productivity, welfare, and farm management efficiency. Wearable devices equipped with sensors can collect physiological, behavioral, and environmental data from animals and transmit this information to farmers in real time. Such technologies enable early detection of diseases, identification of abnormal behaviors, improved reproductive management, and optimization of feeding strategies. Consequently, wearable livestock monitoring systems have become an active area of research and development in modern agriculture.

Several studies have demonstrated the effectiveness of wearable sensor technologies for monitoring animal health and activity. Accelerometers, gyroscopes, temperature sensors, and positioning systems have been widely used to classify animal behaviors such as walking, grazing, standing, resting, and rumination. Research conducted by (Zhang et al., 2021) highlighted the potential of wearable IoT technologies in enabling precision

livestock farming through continuous monitoring of animal conditions. Similarly, (Ding et al., 2025) emphasized the importance of intelligent wearable sensing systems in accurately detecting and analyzing animal behaviors. Advances in machine learning and sensor fusion techniques have further improved the accuracy of behavior recognition and health assessment in livestock production systems.

Among the various sensing technologies available, motion sensors such as accelerometers and gyroscopes have proven particularly useful for monitoring animal activity. These sensors can capture movement patterns that provide valuable information regarding the health and welfare status of livestock. Studies by (Li et al., 2022) and (Pichlbauer et al., 2024) demonstrated that sensor-based monitoring systems can effectively classify animal behaviors and detect deviations from normal activity patterns. Such information can assist farmers in identifying illness, injury, stress, or reproductive events at an early stage, thereby reducing economic losses and improving animal welfare.

In addition to activity monitoring, body temperature serves as an important physiological indicator of animal health. Abnormal temperature variations may signal infections, heat stress, or other health-related conditions. Thermistor-based sensing systems offer a low-cost solution for continuous temperature monitoring and can be integrated into wearable livestock devices. Combined with motion sensing technologies, temperature monitoring provides a more comprehensive assessment of animal well-being and facilitates timely intervention by farmers.

Location tracking is another critical aspect of livestock management, particularly in extensive grazing environments. Global Positioning System (GPS) technology enables farmers to determine the position of animals and monitor movement across grazing areas. GPS-based livestock monitoring systems have been widely adopted to reduce animal losses, improve grazing management, and enhance farm security. However, incorporating a GPS module into every animal collar significantly increases system cost, power consumption, and maintenance requirements. These limitations present challenges for large-scale deployment, especially in developing countries where financial resources may be limited.

Wireless communication technologies play a fundamental role in transmitting sensor data from animals to monitoring stations. Conventional communication technologies such as Wi-Fi and Bluetooth are often limited by range, scalability, and power consumption. In

contrast, emerging communication protocols such as ESP-MESH and LoRa provide promising alternatives for livestock monitoring applications. ESP-MESH enables multiple devices to communicate within a self-organizing network, allowing sensor nodes to relay information efficiently across large areas. LoRa technology offers long-range, low-power communication capabilities that are particularly suitable for remote agricultural environments. Recent studies have demonstrated the effectiveness of LoRa-based systems for livestock monitoring in locations where conventional communication infrastructure is unavailable.

Despite significant progress in wearable livestock monitoring technologies, many existing systems remain expensive due to the requirement of equipping each animal with independent GPS and long-range communication modules. This increases hardware costs, energy consumption, and maintenance complexity, limiting practical adoption among small-scale and medium-scale farmers. Furthermore, scalability becomes a challenge as the number of monitored animals increases. These limitations highlight the need for cost-effective and scalable architectures capable of supporting large livestock populations while maintaining reliable monitoring performance.

This study addresses these challenges through the development of an IoT-based smart wearable device for monitoring the health and activity of farm animals. The proposed system utilizes wearable sensor technologies, ESP32 microcontrollers, ESP-MESH networking, GPS positioning, and LoRa communication to create an integrated livestock monitoring platform. Unlike conventional approaches that require GPS and long-range communication hardware on every collar, the developed architecture employs a shared herdsman gateway that collects data from multiple collar nodes through an ESP-MESH network and subsequently transmits information via LoRa. This approach reduces hardware costs, lowers energy consumption, and improves scalability while maintaining real-time monitoring capabilities.

The developed system aims to provide farmers with accurate information regarding animal movement, activity levels, temperature conditions, and location data. By enabling continuous monitoring and remote data transmission, the proposed solution contributes to improved livestock management, enhanced animal welfare, and the advancement of precision agriculture technologies. Furthermore, the study demonstrates the potential of integrating low-cost IoT technologies into practical agricultural applications, particularly

in regions where affordable and scalable livestock monitoring solutions are needed.

CHAPTER II

LITERATURE REVIEW

The rapid development of digital technologies has significantly transformed agricultural production systems in recent years. Among these advancements, the integration of the Internet of Things (IoT), wireless sensor networks, wearable sensing technologies, and artificial intelligence has enabled the emergence of precision livestock farming. Precision livestock farming focuses on the continuous monitoring of animal health, behavior, welfare, and productivity through automated technologies. These technologies provide farmers with real-time information that supports decision-making and improves overall farm management efficiency.

Traditional livestock monitoring methods rely primarily on manual observation, which can be labor-intensive, subjective, and ineffective for detecting early signs of disease or abnormal behavior. As livestock populations continue to increase globally, automated monitoring systems have become increasingly important. Recent studies have demonstrated that wearable sensors, wireless communication technologies, and intelligent data processing systems can significantly enhance livestock management by providing continuous and accurate information regarding animal conditions.

2.1 Precision Livestock Farming and Animal Monitoring Technologies

Precision livestock farming (PLF) represents a modern approach to animal production that utilizes advanced technologies to monitor individual animals continuously. The objective of PLF is to improve animal welfare, productivity, and farm profitability through data-driven management practices. According to (Besler et al., 2024) precision technologies have become essential tools for monitoring animal behavior, health status, feeding patterns, reproductive performance, and environmental interactions.

Wearable monitoring devices have emerged as one of the most promising technologies in precision livestock farming. These systems are typically attached to the animal's body

and incorporate various sensors capable of measuring movement, physiological parameters, and environmental conditions. A study (Maharajpet et al., 2024) reported that wearable technologies provide farmers with valuable insights into animal health while reducing dependence on manual observation. Similarly, another study (Michelena et al., 2025) emphasized that wearable livestock monitoring systems are becoming increasingly important due to their ability to support real-time decision-making and early disease detection.

2.2 Wearable Sensor Technologies for Livestock Monitoring

Wearable sensors play a critical role in monitoring animal behavior and physiological conditions. Commonly used sensors include accelerometers, gyroscopes, magnetometers, temperature sensors, heart rate sensors, and GPS modules. These sensors collect continuous data that can be analyzed to identify normal and abnormal animal behaviors.

Accelerometer-based monitoring systems are among the most widely adopted wearable technologies in livestock farming. Accelerometers measure acceleration forces along multiple axes and can be used to classify activities such as grazing, standing, walking, running, and resting. Research conducted by (Li et al., 2022) demonstrated that motion sensor data can successfully distinguish between multiple cattle behavioral states with high accuracy.

The MPU9250 sensor combines a three-axis accelerometer, three-axis gyroscope, and three-axis magnetometer into a single compact device. This integrated sensor provides comprehensive motion information while minimizing hardware complexity and power consumption. A study (Ding et al., 2025) highlighted the effectiveness of multi-sensor wearable systems for intelligent animal behavior recognition and monitoring applications.

Temperature monitoring is another important aspect of livestock health management. Abnormal body temperature may indicate infection, fever, stress, or environmental challenges. Thermistor-based sensing systems provide a low-cost solution for continuous temperature monitoring and can be easily integrated into wearable livestock devices.

Sensor Type	Parameter Measured	Application
Accelerometer	Motion	Activity Monitoring
Gyroscope	Rotation	Behaviour Analysis
Magnetometer	Orientation	Movement Tracking
Thermistor	Temperature	Health Monitoring
GPS	Location	Position Tracking

Table 2.1. Common Wearable Sensors Used in Livestock Monitoring

2.3 Internet of Things (IoT) in Livestock Monitoring

The Internet of Things refers to a network of interconnected devices capable of collecting, transmitting, and processing data automatically. IoT technologies have become increasingly important in livestock farming due to their ability to support real-time monitoring and remote management.

According to (Curti et al., 2023), IoT-based livestock monitoring systems enable continuous data collection from animals and facilitate timely interventions by farmers. Similarly, the systematic review conducted by MDPI Sustainability (2024) concluded that IoT technologies improve productivity, reduce labor requirements, and enhance animal welfare.

IoT architectures typically consist of sensing devices, communication networks, gateways, cloud platforms, and user interfaces. Sensor data collected from wearable devices are transmitted through communication networks to a gateway or monitoring station where data can be stored, analyzed, and visualized.

Recent research by (Zhang et al., 2021) demonstrated that wearable IoT systems can successfully support precision livestock farming through continuous monitoring of animal activity and health conditions.

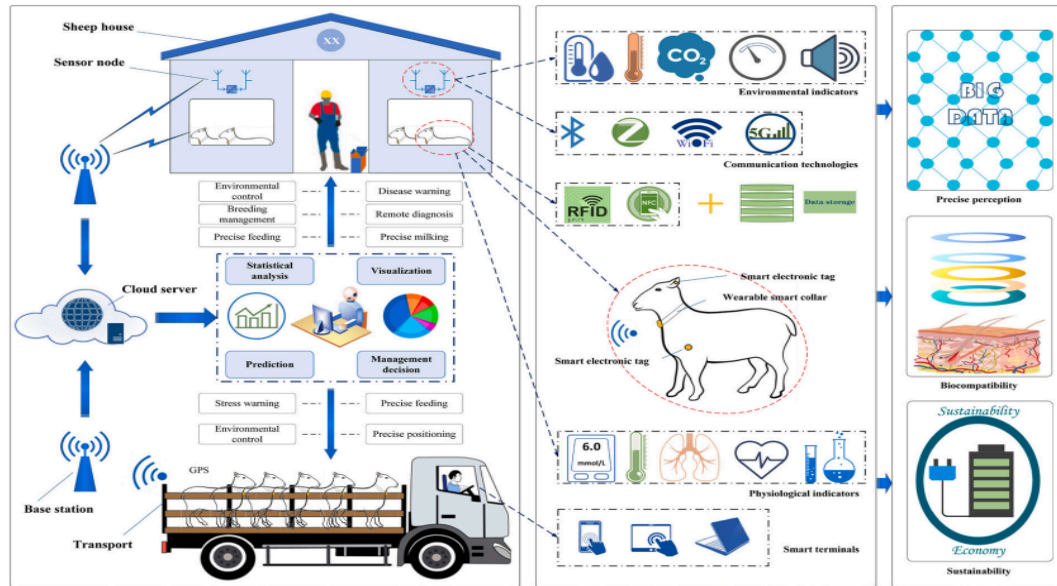


Figure 2.2. General Architecture of an IoT-Based Livestock Monitoring System

2.4 Wireless Communication Technologies in Smart Livestock Farming

Wireless communication forms the backbone of IoT-based livestock monitoring systems. Several communication technologies have been investigated for agricultural applications, including Bluetooth, Wi-Fi, ZigBee, LoRa, and cellular networks.

LoRa (Long Range) technology has gained considerable attention due to its ability to provide long-distance communication while maintaining low power consumption. LoRa networks are particularly suitable for remote livestock production systems where internet infrastructure may be unavailable. Research on LoRa-based livestock monitoring systems has demonstrated communication ranges of several kilometers under open-field conditions.

A LoRa-based cattle monitoring system presented by researchers in 2024 demonstrated reliable communication performance in remote pasture environments while maintaining low energy consumption. Similarly, studies involving LoRa IoT frameworks have shown that LoRa technology can effectively support real-time livestock monitoring in large farming areas.

Technology	Range	Power Consumption	Livestock Suitability
Bluetooth	Short	Low	Low
Wi-Fi	Medium	High	Moderate
ZigBee	Medium	Low	Moderate
LoRa	Long	Low	High

Table 2.2. Comparison of Wireless Communication Technologies

2.5 ESP-MESH Networking for Multi-Collar Systems

ESP-MESH is a networking protocol developed for ESP32 devices that enables multiple nodes to communicate within a self-organizing wireless network. Unlike conventional point-to-point communication systems, mesh networks allow devices to relay information through neighboring nodes, improving network coverage and scalability.

Mesh networking offers significant advantages for livestock monitoring applications. Multiple animal collars can communicate with a single gateway without requiring independent long-range communication modules on every device. This architecture reduces hardware costs, power consumption, and maintenance requirements while increasing system scalability.

The use of mesh networking is particularly attractive for livestock farming because herd sizes can vary significantly. A mesh-based architecture allows additional collars to be incorporated into the network without major modifications to the overall system.

ESP-MESH Architecture for Multi-Collar Livestock Monitoring

Collar 1 → Gateway

Collar 2 → Gateway

Collar 3 → Gateway



GPS + LoRa Gateway



Monitoring Station

2.6 Existing Smart Livestock Monitoring Systems and Research Gaps

Numerous studies have investigated wearable livestock monitoring systems using combinations of sensors, wireless communication technologies, and intelligent data processing techniques. A study (Yu et al., 2024) developed an intelligent wearable device capable of monitoring cattle health in real time using wearable sensors and wireless communication technologies. Their study demonstrated the feasibility of integrating multiple sensing components into a single wearable platform.

Although existing systems have achieved promising results, several limitations remain. Many solutions require each animal collar to contain independent GPS and communication modules, increasing system cost and energy consumption. Furthermore, scalability remains a challenge when monitoring large herds. These limitations can restrict adoption among small-scale farmers and agricultural communities in developing countries.

The present study addresses these gaps by proposing a cost-effective architecture based on ESP-MESH networking and a shared gateway approach. Instead of equipping every collar with GPS and LoRa modules, multiple collar nodes communicate with a centralized gateway responsible for location tracking and long-range communication. This architecture reduces deployment costs while maintaining real-time monitoring capabilities.

Feature	Existing Systems	Proposed System
GPS on Every Collar	Yes	No
LoRa on Every Collar	Often Yes	No
Mesh Networking	Limited	Yes
Shared Gateway	Rare	Yes
Cost Efficiency	Moderate	High
Scalability	Moderate	High

Table 2.3. Comparison Between Existing Systems and Proposed System

This comparison table is very important because it highlights the novelty of the study and prepares the reader for the methodology presented in the following chapter.

CHAPTER III

MATERIALS AND METHODS

This chapter describes the materials, hardware components, software tools, system architecture, communication protocols, and experimental procedures used in the development of the IoT-based smart wearable device for monitoring the health and activity of farm animals. The proposed system was designed to provide real-time monitoring of animal movement, temperature, and location while maintaining low deployment costs and scalability for livestock farming applications.

The developed prototype consists of two main subsystems: the animal collar node and the herdsman gateway node. The collar node is attached to the animal and is responsible for collecting health and activity data. The gateway node receives data from one or more collar nodes through a wireless mesh network, adds location information, and forwards the data over a long-range communication channel to a monitoring station. The overall architecture was designed to reduce hardware costs by eliminating the need for GPS and LoRa modules on every collar while maintaining real-time monitoring capabilities.

3.1 Materials Used in the Study

The hardware components used in the development of the system are presented in Table 3.1.

Component	Quantity	Purpose
ESP32 Development Board	2	Main microcontroller and wireless communication
MPU9250 Sensor Module	2	Motion and activity monitoring
Thermistor Sensor	2	Temperature measurement

NEO-6M GPS Module	1	Location tracking
DX-LR02 SX1278 LoRa Module	2	Long-range communication
USB-to-TTL Converter	1	Lora receiver interface
Rechargeable Battery	2	Portable power supply
Breadboard and Jumper Wires	Several	Prototype development
Personal Computer	1	Data monitoring and testing

Table 3.1. Hardware Components Used in the Developed System

The ESP32 microcontroller was selected due to its integrated Wi-Fi capability, low power consumption, sufficient processing power, and support for ESP-MESH networking. The MPU9250 sensor was used because it combines a three-axis accelerometer, three-axis gyroscope, and three-axis magnetometer in a single compact module. Temperature monitoring was achieved using a thermistor sensor integrated into the collar unit. A NEO-6M GPS module was installed only on the gateway node to reduce system cost and power consumption. Long-range communication between the gateway and monitoring station was implemented using the DX-LR02 SX1278 LoRa modules operating at 433 MHz.

3.2 System Architecture

The developed system consists of two wearable collar nodes and one herdsman gateway node. Each collar node collects animal activity and temperature information and transmits the data through an ESP-MESH network. The gateway node receives the information from all connected collars, appends GPS location data, and forwards the complete data packet to a remote monitoring station through LoRa communication.

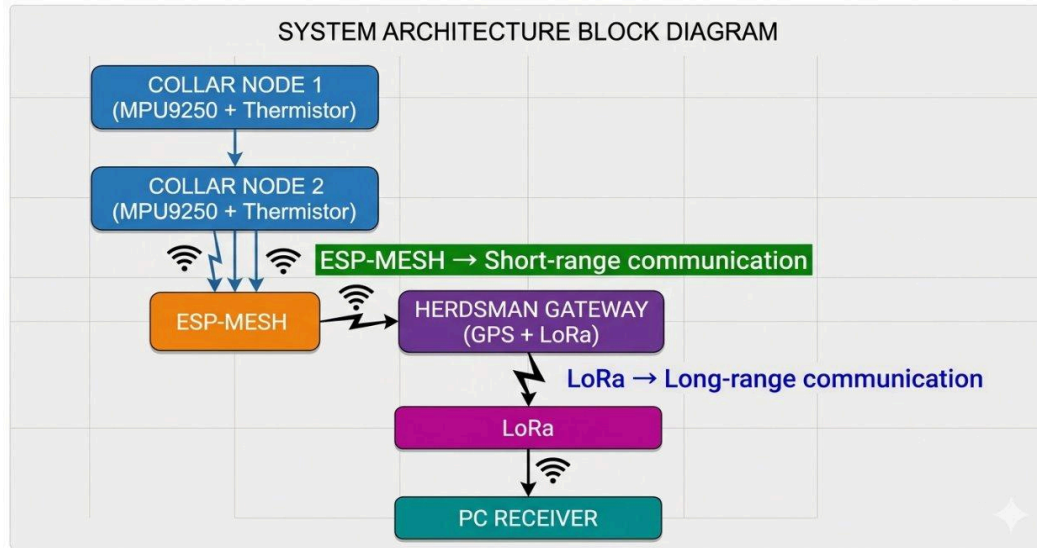


Figure 3.1. Overall System Architecture

The architecture was designed to minimize hardware redundancy. Instead of installing GPS and LoRa modules on every collar, a shared gateway architecture was adopted. This approach reduces hardware cost, decreases power consumption, and improves scalability when additional collar nodes are introduced into the system.

3.3 Collar Node Design

The collar node was designed as a wearable device attached to the animal's neck. It consists of an ESP32 microcontroller, MPU9250 motion sensor, thermistor temperature sensor, and rechargeable battery.

The MPU9250 sensor continuously measures acceleration, angular velocity, and orientation data. These measurements are used to evaluate animal movement and activity levels. The thermistor sensor monitors temperature variations that may indicate potential health abnormalities. Sensor readings are processed by the ESP32 and transmitted to the gateway through ESP-MESH communication.

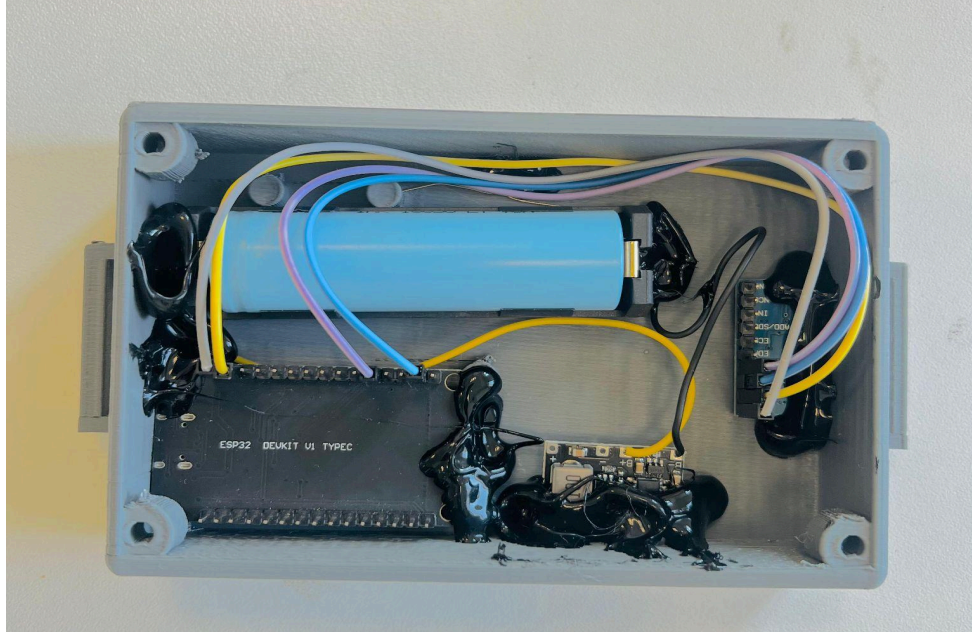


Figure 3.2. Collar Node Hardware Design

Collar Node Pin Connections

Component	ESP32 Pin
MPU9250 SDA	GPIO 21
MPU9250 SCL	GPIO 22
Thermistor Output	GPIO 34
Battery Supply	VIN
Ground	GND

Table 3.2. Collar Node ESP32 Connections

3.4 Herdsman Gateway Design

The gateway node acts as the central communication hub of the system. It receives

information from multiple collar nodes through ESP-MESH networking and transmits the collected data to a monitoring station using LoRa communication.

The gateway consists of an ESP32 microcontroller, NEO-6M GPS module, DX-LR02 LoRa module, and rechargeable battery. The GPS module provides location information that can be associated with the animal data before transmission.

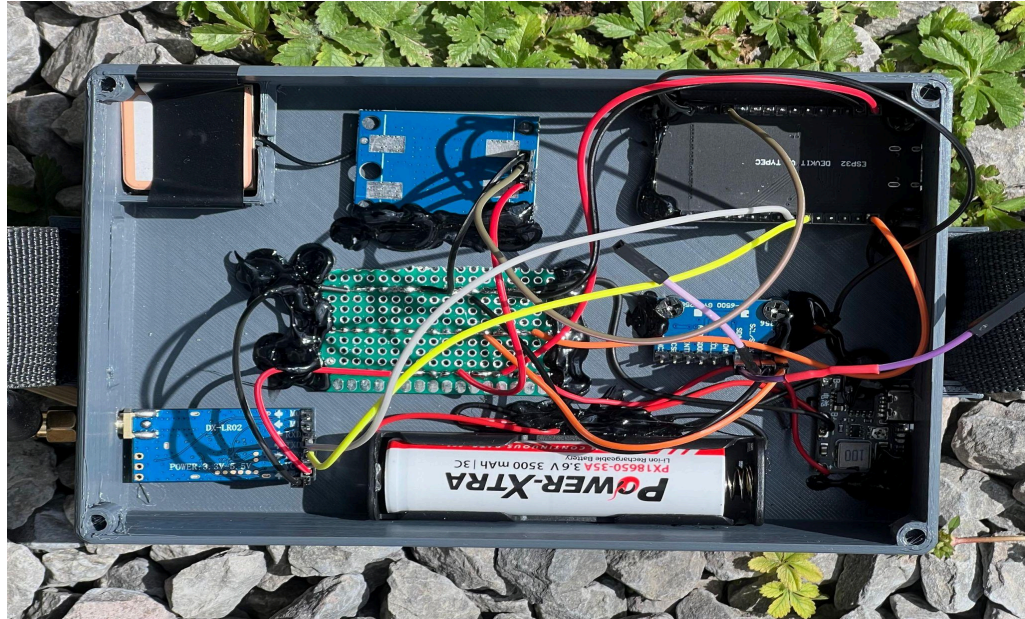


Figure 3.3. Herdsman Gateway Hardware Design

Gateway Pin Connections

Component	ESP Pin
GPS TX	GPIO 32
GPS RX	Not Used
LoRa TX	GPIO 16
LoRa RX	GPIO 17
Battery Supply	VIN
Ground	GND

Table 3.3. Gateway ESP32 Connections

3.5 ESP-MESH Communication Network

Communication between the collar nodes and the gateway was established using ESP-MESH technology. ESP-MESH is a self-organizing wireless networking protocol developed for ESP32 devices. The protocol enables multiple devices to communicate through a distributed network without requiring traditional Wi-Fi infrastructure.

In the developed system, each collar node functions as a mesh node, while the herdsman device serves as the root gateway. Sensor information collected from the collars is transmitted to the gateway through the mesh network.

The ESP-MESH architecture provides several advantages including scalability, low power consumption, self-healing network capabilities, and support for multiple collar nodes. Additional collars can be incorporated into the network without significant modifications to the existing architecture.

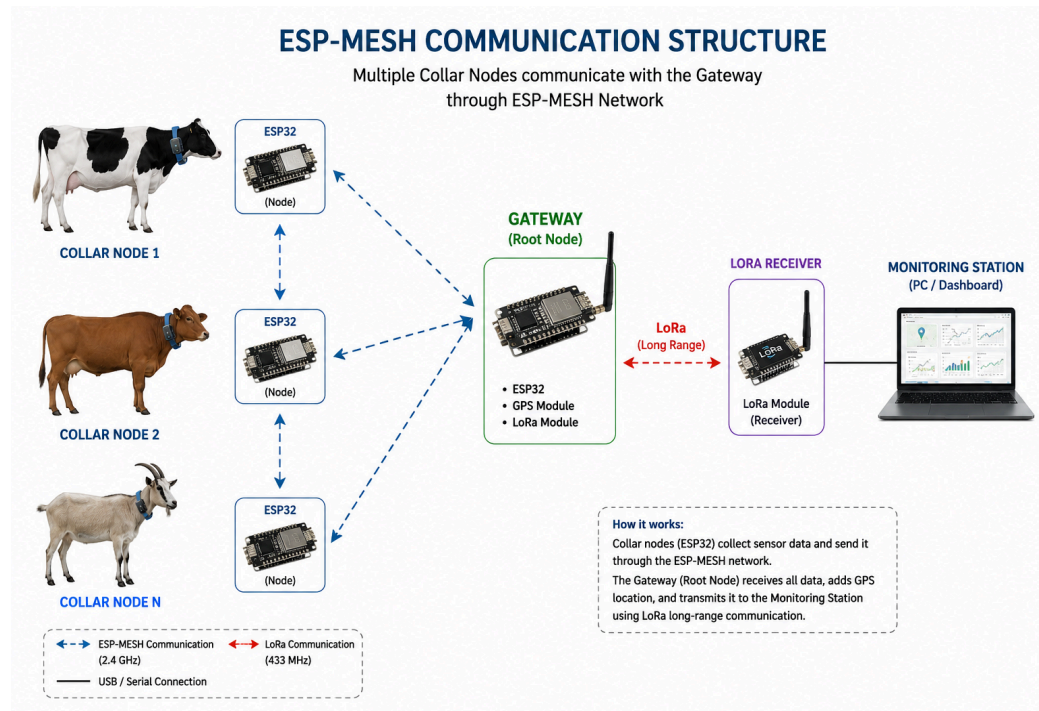


Figure 3.4. ESP-MESH Communication Structure

3.6 LoRa Communication System

Long-range communication between the gateway and monitoring station was achieved using SX1278 LoRa modules operating at 433 MHz. LoRa technology was selected due to its low power consumption and long communication range, making it suitable for livestock monitoring applications in remote agricultural environments.

The gateway transmits the collected sensor information to a LoRa receiver connected to a personal computer through a USB-to-TTL converter. Experimental testing indicated that LoRa communication can operate over distances approaching 3 km under favorable environmental conditions.

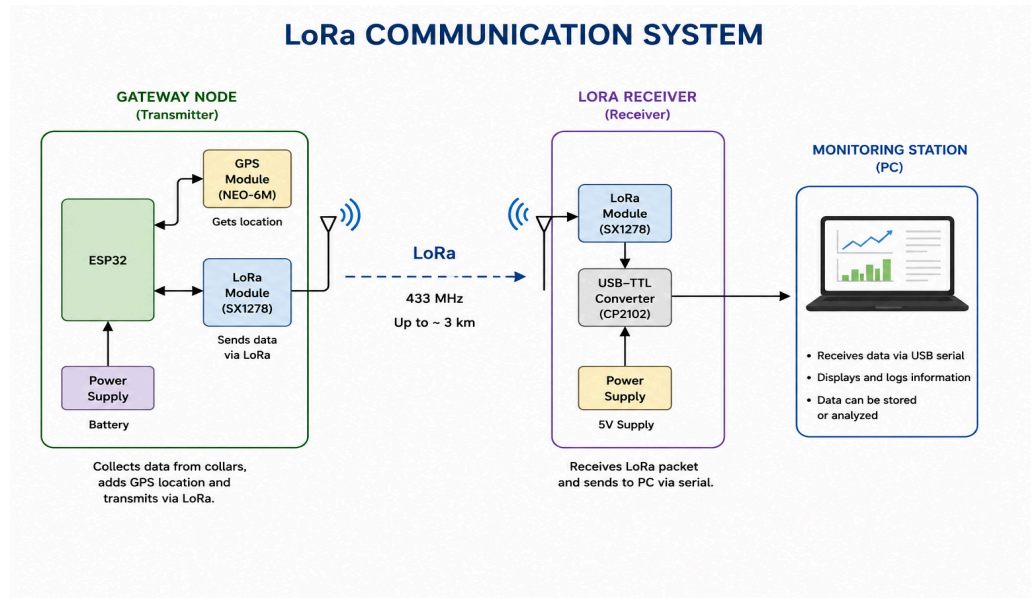


Figure 3.5. LoRa Communication System

3.7 Software Development

The software for the developed system was implemented using the Arduino Integrated

Development Environment (Arduino IDE). Several libraries were incorporated to support sensor communication, GPS data processing, and wireless networking.

Library	Purpose
Wire.h	I ² C Communication
TinyGPSPlus.h	GPS Data Processing
SparkFunMPU9250-DMP.h	MPU9250 Sensor Control
painlessMesh.h	ESP-Mesh Networking
WiFi.h	ESP32 Wireless Functions

Table 3.4. Software Libraries Used

The collar node software continuously acquires sensor measurements, processes the data, and transmits the information through the mesh network. The gateway software receives incoming packets, acquires GPS coordinates, combines the information into a single data packet, and forwards the result via LoRa communication.

An example transmitted data packet is shown below:

COLLAR_1, TEMP=31.20, AX=0.25, AY=0.71, AZ=-0.65, LAT=37.966541, LON=34.679233

3.8 Experimental Testing Procedure

The developed prototype was assembled using breadboard-based hardware connections and tested under laboratory and outdoor conditions. Individual components including the MPU9250 sensor, GPS module, ESP-MESH network, and LoRa communication system were first validated independently. Following successful individual testing, the components were integrated into the complete monitoring system.

The experimental evaluation focused on verifying sensor data acquisition, ESP-MESH communication reliability, GPS location acquisition, LoRa data transmission, and overall system functionality. Sensor readings received at the monitoring station were observed

through the serial monitor interface to confirm successful end-to-end communication between collar nodes and the remote receiver.

The methodology adopted in this study enabled the successful development and testing of a low-cost, scalable, and IoT-enabled livestock monitoring system suitable for modern precision livestock farming applications.

CHAPTER IV

RESULTS AND DISCUSSIONS

This chapter presents the results obtained from the implementation of the IoT-based smart wearable livestock monitoring system and discusses the performance of the developed prototype. The system was evaluated based on sensor data acquisition, wireless communication reliability, and overall end-to-end data transmission from the animal collar nodes to the remote monitoring station.

4.1 System Functionality Results

The developed system successfully demonstrated real-time monitoring of livestock health and activity parameters. The collar nodes were able to collect motion and temperature data using the MPU9250 sensor and thermistor, respectively. These readings were processed by the ESP32 microcontroller and transmitted through the ESP-MESH network to the herdsman gateway.

The gateway node successfully received data packets from the collar nodes and appended GPS location information before forwarding the combined data via LoRa communication to the monitoring station. This confirms that the multi-layer communication architecture (ESP-MESH + LoRa) functions as intended.

Building sketch Ln 91, Col 1 DOIT ESP32 DEVKIT V1 on COM3 2

```
17:51:04.359 -> 154,COLLAR_1,TEMP=0.00,AX=0.039,AY=-0.003,AZ=-0.991,GPS_WAITING_FOR_SIGNAL
17:51:07.976 -> 155,COLLAR_1,TEMP=0.00,AX=0.039,AY=-0.006,AZ=-0.984,GPS_WAITING_FOR_SIGNAL
17:51:11.571 -> 156,COLLAR_1,TEMP=0.00,AX=0.045,AY=-0.011,AZ=-0.987,GPS_WAITING_FOR_SIGNAL
17:51:15.198 -> 158,COLLAR_1,TEMP=0.00,AX=0.046,AY=-0.012,AZ=-1.000,GPS_WAITING_FOR_SIGNAL
17:51:18.816 -> 160,COLLAR_1,TEMP=0.00,AX=0.034,AY=-0.008,AZ=-0.990,GPS_WAITING_FOR_SIGNAL
17:51:22.399 -> 162,COLLAR_1,TEMP=0.00,AX=0.050,AY=-0.021,AZ=-1.002,GPS_WAITING_FOR_SIGNAL
17:51:26.039 -> 163,COLLAR_1,TEMP=0.00,AX=0.042,AY=-0.013,AZ=-0.997,GPS_WAITING_FOR_SIGNAL
17:51:29.645 -> 165,COLLAR_1,TEMP=0.00,AX=0.042,AY=0.538,AZ=-0.818,GPS_WAITING_FOR_SIGNAL
17:51:33.241 -> 167,COLLAR_1,TEMP=0.00,AX=0.046,AY=-0.008,AZ=-0.994,GPS_WAITING_FOR_SIGNAL
```

Figure 4.1. Sample received data at the monitoring station

```
DOIT ESP32 DEVKIT V1' on 'COM3')

TEMP=0.00,AX=0.041,AY=-0.013,AZ=-0.981

TEMP=0.00,AX=0.044,AY=-0.007,AZ=-0.998

TEMP=0.00,AX=0.051,AY=-0.005,AZ=-0.984

TEMP=0.00,AX=0.050,AY=-0.008,AZ=-0.980

TEMP=0.00,AX=0.039,AY=-0.015,AZ=-0.988

TEMP=0.00,AX=0.038,AY=-0.007,AZ=-0.984

TEMP=0.00,AX=0.040,AY=-0.016,AZ=-0.987

TEMP=0.00,AX=0.041,AY=-0.015,AZ=-0.975

TEMP=0.00,AX=0.045,AY=-0.005,AZ=-1.000
```

Ln 91, Col 1

Figure 4.2. Sample received data at the collar node

```
34 // Send via LoRa
35 LORA_SERIAL.println(finalPacket);
```

Serial Monitor X

Message (Enter to send message to 'DOIT ESP32 DEVKIT V1' on 'COM7')

17:50:39.645 -> GPS connected but waiting for satellites...
17:50:39.645 -> This is NORMAL indoors.
17:50:39.645 -> Satellites Found: 0
17:50:39.645 -> =====
17:50:40.809 -> =====
17:50:40.809 -> Received from Node: 476545245
17:50:40.809 -> 206,COLLAR_1,TEMP=0.00,AX=0.045,AY=-0.013,AZ=-0.985
17:50:40.809 -> LoRa Sent:
17:50:40.809 -> 206,COLLAR_1,TEMP=0.00,AX=0.045,AY=-0.013,AZ=-0.985,GPS_WAITING_FOR_SIGNAL
17:50:42.745 -> =====
17:50:42.745 -> Received from Node: 476545245
17:50:42.745 -> 207,COLLAR_1,TEMP=0.00,AX=0.042,AY=-0.010,AZ=-0.975
17:50:42.786 -> LoRa Sent:
17:50:42.786 -> 207,COLLAR_1,TEMP=0.00,AX=0.042,AY=-0.010,AZ=-0.975,GPS_WAITING_FOR_SIGNAL

Ln 31, Col 37

Figure 4.3. Sample received data at the gateway node

4.2 Sensor Performance Results

The MPU9250 sensor provided stable accelerometer and gyroscope readings during both stationary and movement conditions of the prototype collar. The thermistor sensor also responded effectively to temperature changes, indicating its suitability for basic livestock temperature monitoring.

Although the system does not include advanced calibration for clinical temperature measurement, the sensor outputs were sufficient for detecting relative changes in body temperature trends, which is useful for early anomaly detection.

4.3 Communication Performance Results

ESP-MESH Communication

The ESP-MESH network successfully connected multiple collar nodes to the gateway without requiring a fixed Wi-Fi infrastructure. Data transmission between nodes showed low delay under short-range testing conditions. However, minor buffering delays were observed when multiple packets were transmitted simultaneously.

LoRa Communication

The LoRa module achieved reliable long-range communication between the gateway and the monitoring station. The system demonstrated stable transmission up to approximately 3 km under open-field conditions. This confirms LoRa as a suitable technology for rural and remote livestock monitoring applications.

4.4 System Limitations

During testing, some limitations were observed:

- GPS signal acquisition was weak indoors, leading to missing or zero coordinates
- Slight delay occurred when multiple collar nodes transmitted data simultaneously
- MPU9250 initialization occasionally failed if wiring was loose or unstable
- LoRa transmission experienced minor buffering delays during continuous sending

Despite these limitations, the system maintained overall functional integrity and successful end-to-end communication.

4.5 Discussion

The results demonstrate that the proposed system is effective for real-time livestock monitoring using a low-cost IoT approach. The combination of ESP-MESH and LoRa provided a scalable and efficient communication framework. By centralizing GPS and LoRa at the gateway node, the system significantly reduces hardware cost per animal collar, making it more suitable for large-scale livestock deployment.

Compared to traditional livestock tracking systems that require individual GPS-enabled collars, this architecture provides a more cost-effective and energy-efficient solution. The modular design also allows future expansion, including additional sensor integration and intelligent data processing using machine learning techniques.

Overall, the system fulfills the primary objective of enabling real-time monitoring of animal health and activity with reliable wireless communication and scalable architecture suitable for precision livestock farming applications.

CHAPTER V

CONCLUSION

5.1 Conclusion

This study successfully developed an IoT-based smart wearable device for monitoring the health and activity of farm animals. The system integrated ESP32 microcontrollers, MPU9250 motion sensors, thermistor temperature sensors, GPS technology, ESP-MESH networking, and LoRa communication to create a real-time livestock monitoring platform.

The developed architecture consisted of animal collar nodes and a herdsman gateway node. The collar nodes were responsible for collecting motion and temperature data, while the gateway node received information through ESP-MESH communication, added location information using GPS, and transmitted the data to a monitoring station via LoRa. The prototype demonstrated successful acquisition, processing, and wireless transmission of livestock monitoring data.

The use of ESP-MESH networking enabled communication between multiple collar nodes and a single gateway, while the LoRa module provided long-range communication suitable for remote farming environments. By placing the GPS and LoRa modules only at the gateway node rather than on every collar, the proposed system significantly reduced hardware cost, power consumption, and overall system complexity.

The results obtained from the experimental testing confirmed that the developed system can provide real-time monitoring of livestock activity and health-related parameters. Therefore, the proposed system offers a low-cost, scalable, and practical solution for precision livestock farming and has the potential to improve animal welfare, farm management efficiency, and early detection of abnormal animal conditions.

5.2 Recommendations

Based on the findings of this study, the following recommendations are proposed for future work:

- Integrate machine learning and artificial intelligence algorithms for automatic detection of abnormal animal behavior and potential health issues.
- Develop a mobile application and web dashboard to enable real-time visualization and remote monitoring of livestock data.
- Implement cloud-based data storage to support long-term analysis and historical record keeping.
- Increase the number of collar nodes in the ESP-MESH network to evaluate system performance under larger herd conditions.
- Incorporate additional sensors such as heart rate, humidity, and environmental sensors to improve livestock health assessment.
- Design a custom printed circuit board (PCB) and compact enclosure to improve durability, portability, and commercial viability.
- Explore the use of solar-powered charging systems to extend operational lifetime in outdoor farming environments.
- Implement real-time alert and notification systems that can automatically notify farmers when abnormal temperature, inactivity, or unusual movement patterns are detected.

The implementation of these recommendations would further enhance the functionality, reliability, and applicability of the developed smart wearable livestock monitoring system for modern precision agriculture.

REFERENCES

- Besler, B. C., Mojabi, P., Lasemiimani, Z., Murphy, J. E., Wang, Z., Baker, R., ... & Fear, E. C. (2024). Scoping review of precision technologies for cattle monitoring. *Smart Agricultural Technology*, 9, 100596.
- Curti, P. D. F., Selli, A., Pinto, D. L., Merlos-Ruiz, A., Balieiro, J. C. D. C., & Ventura, R. V. (2023). Applications of livestock monitoring devices and machine learning algorithms in animal production and reproduction: an overview. *Animal reproduction*, 20(2), e20230077.
- Ding, L., Zhang, C., Yue, Y., Yao, C., Li, Z., Hu, Y., ... & Li, Q. (2025). Wearable sensors-based intelligent sensing and application of animal behaviors: A comprehensive review. *Sensors*, 25(14), 4515.
- Li, Y., Shu, H., Bindelle, J., Xu, B., Zhang, W., Jin, Z., ... & Wang, W. (2022). Classification and analysis of multiple cattle unitary behaviors and movements based on machine learning methods. *Animals* 12 (9), 1060.
- Michelena, Á., Fontenla-Romero, Ó., & Luis Calvo-Rolle, J. (2025). A review and future trends of precision livestock over dairy and beef cow cattle with artificial intelligence. *Logic Journal of the IGPL*, 33(4), jzae111.
- Pichlbauer, B., Chapa Gonzalez, J. M., Bobal, M., Guse, C., Iwersen, M., & Drillich, M. (2024). Evaluation of different sensor systems for classifying the behavior of dairy cows on pasture. *Sensors*, 24(23), 7739.
- Zhang, M., Wang, X., Feng, H., Huang, Q., Xiao, X., & Zhang, X. (2021). Wearable Internet of Things enabled precision livestock farming in smart farms: A review of technical solutions for precise perception, biocompatibility, and sustainability monitoring. *Journal of Cleaner Production*, 312, 127712.

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